

Beyond the Formal Grid: The Role of Panchayat Expenditure and Infrastructure Quality on Rural Unorganised Firm Performance

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Abstract

The unincorporated non-agricultural sector plays a critical role in the Indian economy by generating employment, supporting livelihoods, and providing essential linkages to the formal sector. Using data from the Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2023–24, this study examines the determinants of unincorporated enterprise performance in rural India. The analysis is supplemented with district-level data on rural development expenditure (2022–24) and infrastructure indicators from the Mission Antyodaya Survey 2022–23. Owing to data-matching constraints, the sample is restricted to proprietary enterprises across 21 states. Descriptive findings indicate that while enterprises and workers are distributed relatively evenly across manufacturing, trade, and other services, gross value added (GVA) is disproportionately concentrated in the other services sector. The study further documents variations in enterprise outcomes across enterprise age, location, industry, and owner characteristics. To identify the determinants of performance, regression models are estimated using the logarithm of real GVA as the dependent variable and enterprise and owner characteristics as explanatory variables. The results highlight the importance of infrastructure, human capital, and enterprise-specific factors in shaping performance outcomes. The findings contribute to a better understanding of the rural unincorporated sector and provide evidence for designing more targeted and nuanced policy interventions aimed at improving productivity and enterprise growth.

Keywords: Rural Enterprises; Unincorporated Sector; Infrastructure; Entrepreneurship; Firm Performance; Informal Sector; Unorganised Sector

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I. Motivation

The unincorporated non-agricultural sector occupies an important place in the Indian economy. It contributes substantially to employment, output, and the broader socio-economic landscape³, particularly in rural areas where formal wage employment opportunities remain limited. The sector includes a wide range of activities across manufacturing, trade, and other services, and provides livelihood opportunities to low-skilled workers, household-based producers, own-account workers⁴, and small entrepreneurs. In many cases, these enterprises also support local supply chains, preserve indigenous forms of production, and act as input suppliers or service providers to other sectors of the economy.

The importance of this sector is also visible in the continued policy attention given to entrepreneurship, self-employment, credit access, and small enterprise development. Over the years, the government has introduced several schemes aimed at supporting enterprise creation and expansion, including programmes such as Stand-Up India, Startup India, PM Vishwakarma, PM Svanidhi, MUDRA, and other Micro, Small and Medium Enterprises⁵ (MSME)-oriented initiatives. These programmes primarily focus on promoting entrepreneurship, facilitating access to credit, supporting enterprise formalisation, enhancing skills and productivity, and strengthening the growth prospects of micro and small businesses. The frequent announcement and extension of such schemes also indicates that this sector remains central to India's development strategy.

However, the need for sustained policy attention also points to the continuing constraints faced by these enterprises⁶. Despite their scale and importance, many unincorporated enterprises operate with low capital, limited access to formal finance, weak market linkages, low technology adoption, and inadequate infrastructure. Poor local infrastructure can restrict the ability of enterprises to procure inputs, reach consumers, adopt technology, or expand beyond a subsistence scale.

Recent evidence from the Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2023-24 further highlights the persistence of these structural constraints. Despite a recovery in enterprise numbers, employment, and gross value added following the pandemic, the sector continues to be dominated by own-account enterprises and very small firms with low productivity, limited access to formal credit, and modest levels of digital adoption. The evidence suggests that the challenge facing the unincorporated sector is not merely one of enterprise creation, but also of enterprise growth and productivity enhancement⁷.

A further challenge is that the unincorporated sector is often discussed within the broader MSME framework. Many rural unincorporated enterprises are extremely small and household-based, and may be better understood as nano-enterprises⁸ rather than conventional micro enterprises. These units may have a low fixed capital, few or no hired workers, limited access to formal credit, and weak integration with wider markets. As a result, their constraints may differ from those faced by larger micro and small enterprises that are more visible within formal MSME policy frameworks. Treating the sector as a single category can therefore lead to an incomplete understanding of how these enterprises operate.

Against this background, this study examines the structure and performance of the rural unincorporated sector using data from ASUSE 2023–24. The rural focus is important because rural unincorporated enterprises operate in a different environment from urban enterprises. The study documents descriptive

³ [MoSPI, 2024](#)

⁴ Own account worker is defined as one person on their own, or along with some partners, running business without hiring any paid labour.

⁵ As per the MSME classification (2020), enterprises are categorized as micro, small, or medium based on combined thresholds of investment in plant and machinery/equipment and annual turnover. The respective limits are ₹1 crore and ₹5 crore for micro enterprises, ₹10 crore and ₹50 crore for small enterprises, and ₹50 crore and ₹250 crore for medium enterprises.

⁶ Enterprises and firms are used interchangeably in this paper.

⁷ Gautam, & Dhanya, V. (2026)

⁸ Nano-enterprise is typically managed by a self-employed worker or informal entrepreneur engaged in very small-scale, low-capital business activities. These individuals often lack formal registration, operate in marginalized or rural contexts, and rely on household labour, limited infrastructure, and community-based resources to sustain their ventures (Ramírez-López et al., 2025).

patterns in enterprise activity across sectors, states, industries, skill groups, and age cohorts. It then analyses the factors associated with enterprise-level real gross value added, focusing on enterprise characteristics, owner characteristics, and district-level conditions. Particular attention is given to the role of infrastructure, quality of governance, and changes in government expenditure. By doing so, the study seeks to capture the heterogeneity within the rural unincorporated sector rather than treating it as a homogeneous category of small enterprises.

II. Literature Review

A large body of literature has examined the determinants of firm and industrial performance, focusing on factors ranging from firm characteristics and entrepreneurial attributes to infrastructure, access to finance and institutional conditions. Given the important role of enterprises in generating output, employment and economic growth, understanding the factors that influence enterprise performance remains a key area of research. Existing studies identify several themes that are particularly relevant for explaining variations in firm performance, defined as value added, and productivity parameters in various contexts.

- a. **Firm Activity:** Firm size and age are important determinants of enterprise performance. While large firms tend to be more productive, job creation and employment growth are often concentrated among young firms. Small firms account for a large share of new jobs, although much of this effect reflects their younger age profile (Ács *et al.*, 2006; Ayyagari *et al.*, 2011; Haltiwanger *et al.*, 2013; Decker *et al.*, 2014). However, young firms are more sensitive to business cycles than older and larger firms (Fort *et al.*, 2013). Employment fluctuations among startups are procyclical, and conditions at the time of firm birth explain much of the cohort-level variation in outcomes (Pugsley and Sahin, 2015; Sedlacek, 2017). Hazard rates also decline with firm age, and larger firms tend to survive longer, partly because low-productivity firms exit early (Hopenhayn, 1992).
- b. **Infrastructure/Fiscal Support:** Roads facilitate the movement of workers out of agriculture, without major changes in agricultural outcomes (Asher and Novosad, 2020). Public investment and road creation have a strong positive association with productivity in the formal sector, but no systematic association with the output of the average informal firm (Klapper *et al.*, 2006; Chatterjee *et al.*, 2019). However, PMGSY has been found to positively affect rural informal firms, with no evidence of an effect on formal enterprises (Nandwani *et al.*, 2025). Other studies also show that new roads can support small business creation, as all-weather roads reduce seasonal disruptions and improve access to nearby markets (Agarwal *et al.*, 2022; Sharma *et al.*, 2022; Naaraayanan *et al.*, 2024).
- c. **Unorganised Sector:** The literature on the unorganised sector highlights the role of regulations, informality and firm size. Costly regulations can discourage the creation of new firms (Klapper *et al.*, 2006). Informal enterprises avoid registration costs, taxes and regulatory requirements, which may allow them to save resources and continue operating (La Porta and Shleifer, 2014; Williams *et al.*, 2016). However, the central constraint in India's unorganised sector is not firm creation alone, but firm growth. The sector is largely made up of own-account enterprises and is marked by low productivity, limited credit access and low digital adoption (Gautam and Dhanya, 2026). Informal firms are usually very small, are often managed by less educated owners and rarely transition into formal firms (La Porta and Shleifer, 2008). As countries grow and develop, the informal economy tends to shrink and the formal economy becomes more dominant (La Porta and Shleifer, 2014).
- d. **Credit Constraints:** Institutional credit helps promote rural entrepreneurship and has a positive effect on firm growth (Agarwal *et al.*, 2022; Natarajan *et al.*, 2022; Vidyarani and Mallick, 2025). Access to finance allows enterprises to invest, expand operations and manage risks. Recent evidence also suggests that electronic payments can reduce information asymmetries and improve lending opportunities for small firms (Galilea *et al.*, 2026).

- e. **Female Entrepreneurship:** Women entrepreneurs in India's rural informal sector are emerging, but their enterprises are generally smaller, more home-based and less productive. This reflects limited access to finance, markets, networks and suitable opportunities for women (Ghani *et al.*, 2013; Natarajan *et al.*, 2022; Agarwal *et al.*, 2024).

Overall, the literature suggests that enterprise performance is shaped by both internal firm-level factors and external local conditions. Firm age, size, credit access and ownership characteristics influence productivity and survival, while infrastructure and institutional conditions affect market access and business opportunities. This provides the basis for examining enterprise performance using both enterprise-level and district-level variables.

While previous studies have examined various dimensions of enterprise performance, India's unincorporated sector has received comparatively less attention, especially in studies based on the ASUSE dataset. Given the sector's substantial contribution to employment and economic activity, this represents an important gap in the literature. The present study contributes to addressing this gap by providing empirical evidence from the ASUSE survey and linking the dataset with administrative datasets to examine how local conditions relate to enterprise performance.

III. Data

The Annual Survey of Unincorporated Sector Enterprises (ASUSE) is a nationally representative survey of unincorporated establishments engaged in manufacturing, trade, and other services (excluding construction)⁹. This study uses cross-sectional data for the rural unincorporated sector from the 2023–24 round of the survey¹⁰. In 2023–24, ASUSE estimated approximately 7.34 crore establishments employing 12.06 crore workers across India.

In addition to ASUSE, the analysis incorporates administrative datasets on district-level committed developmental expenditure for the rural sector spanning 2022–2024, along with village-level infrastructure from the 2022–23 Mission Antyodaya survey and governance indicators from Panchayat Advancement Index 2022-23, aggregated to the district level.

To ensure consistency in merging external administrative datasets, the analysis excludes Union Territories and seven states, restricting the sample to 21 states. The study further focuses on proprietary enterprises, yielding an analytical sample representing approximately 3.52 crore establishments and 5.11 crore workers.

IV. Stylised Facts

This section presents a set of stylised facts on the structure of the rural unincorporated sector. These patterns describe the distribution of establishments, employment, output, and fixed capital¹¹ across sectors, states, industries, and firm cohorts (based on age of firm).

1. Overview

⁹ The establishments that are neither registered under Companies Act, 1956 nor under Companies Act, 2013 are only covered.

¹⁰ The unit of enquiry of the ASUSE is an Establishment.

¹¹ Fixed Capital and Investment imply the same variable in this paper.

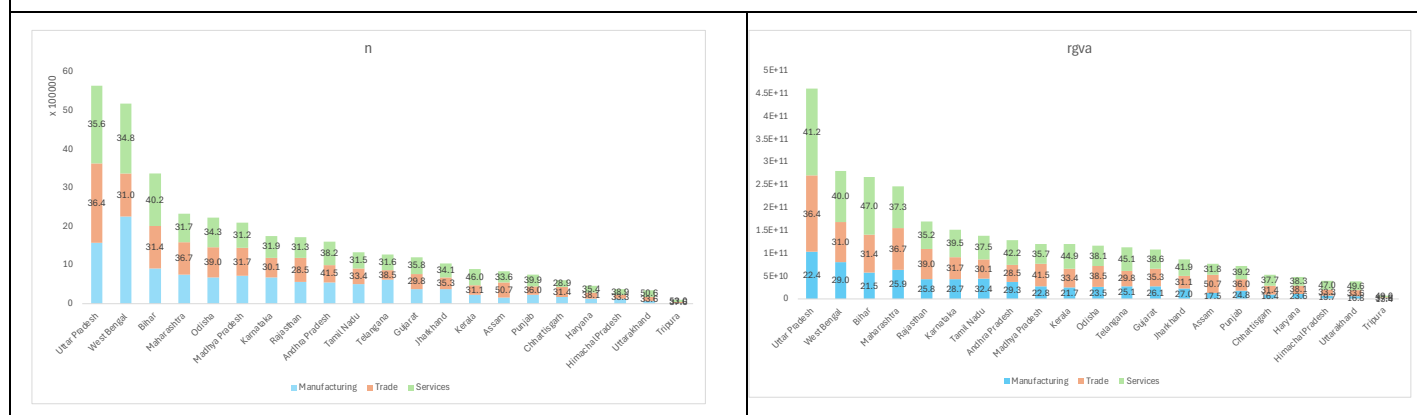
Table 1: Data Overview

		All	3-sector		
S.N		All-India	Manufacturing	Trade	Other Services
1.	Count (In Crore)	3.5	1.2	1.1	1.2
2.	Workers (In Crore)	5.1	1.7	1.7	1.7
3.	GVA ¹² (In ₹ Lakh Crore)	28.7	7.1	9.9	11.7
4.	Fixed Capital ¹³ (In ₹ Lakh Crore)	62.8	17.2	20.9	24.7
5.	To/Ko ¹⁴ (median)	1.3	0.6	3.7	1.3

Employment is relatively balanced across the three sectors. However, other services account for the largest share of aggregate GVA despite employing a similar number of workers, suggesting higher average productivity relative to manufacturing and trade.

2. State-Level Profile

Figure 1: State Level Profile [Clockwise: Number of Establishments, GVA Contribution (in ₹), GVA per establishment (in ₹) and Investment per establishment (in ₹)]

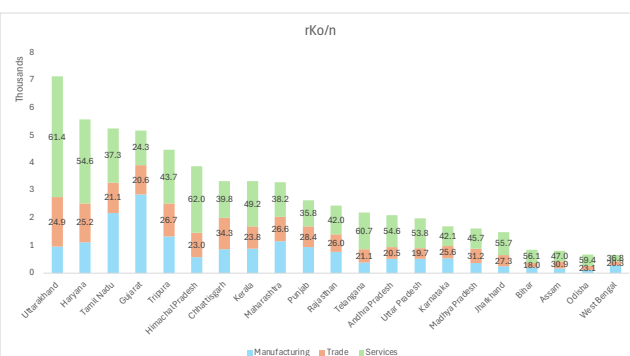
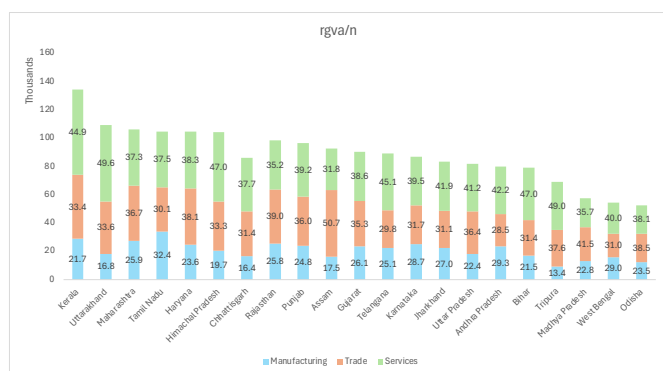


¹² All financial values (GVA, Fixed Cap, Turnover, Expenditure etc.) used in the paper are in real terms, using CPI 2023-24.

¹³ Fixed Capital includes value of owned land, building, machinery, tools and other equipment, ICT equipment etc.

¹⁴ Stands for Turnover/Fixed Cap ratio.

Figure 1: State Level Profile [Clockwise: Number of Establishments, GVA Contribution (in ₹), GVA per establishment (in ₹) and Investment per establishment (in ₹)]



Note: All values are in real terms.¹⁵

States differ considerably in the scale and characteristics of rural enterprise activity. For example, Uttar Pradesh accounts for a large concentration of establishments and aggregate GVA, yet this is not accompanied by proportionately higher GVA or fixed capital per establishment. In contrast, some states with a smaller enterprise¹⁶ base exhibit stronger enterprise-level outcomes, indicating that differences across states are shaped not only by the scale of activity, but also by the underlying composition of enterprises.

3. Top Industries

Table 2: Top 5 Industries

S.N.	Industry	GVA (%)	Employment (%)	Establishments (%)	Fixed Capital (%)
1.	Retail Trade, except of Motor Vehicles and Motorcycles	29.3	30.1	28.5	29.6
2.	Land Transport	14.0	8.7	11.7	12.8
3.	Manufacture of Wearing Apparel	6.0	11.3	15.0	9.7
4.	Food and Beverage Service Activities	7.3	6.5	4.8	4.9
5.	Manufacture of Food Products	4.7	4.7	3.7	5.4
	Total share of 5 industries.	61.4	61.3	63.8	62.3

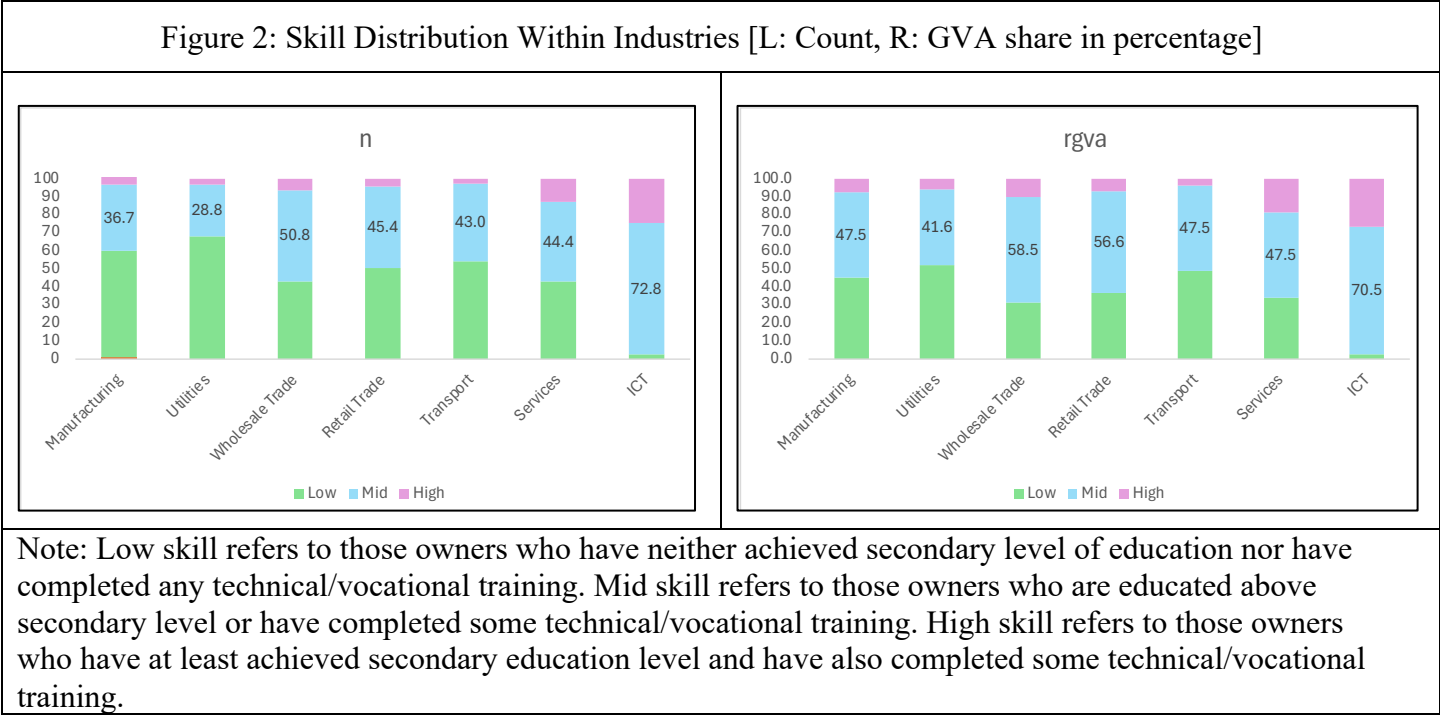
A small number of industries account for a disproportionately large share of enterprise activity, with only five industries (out of a total of 68 industries) contributing over 60 percent of aggregate GVA, employment, establishments, and fixed capital, with retail trade dominating across all indicators. The

¹⁵ Annexes available [online](#).

¹⁶ The enterprise and the establishment are the same for single-establishment firms. Since most of the informal sector operates from a single location, they are used interchangeably in this study.

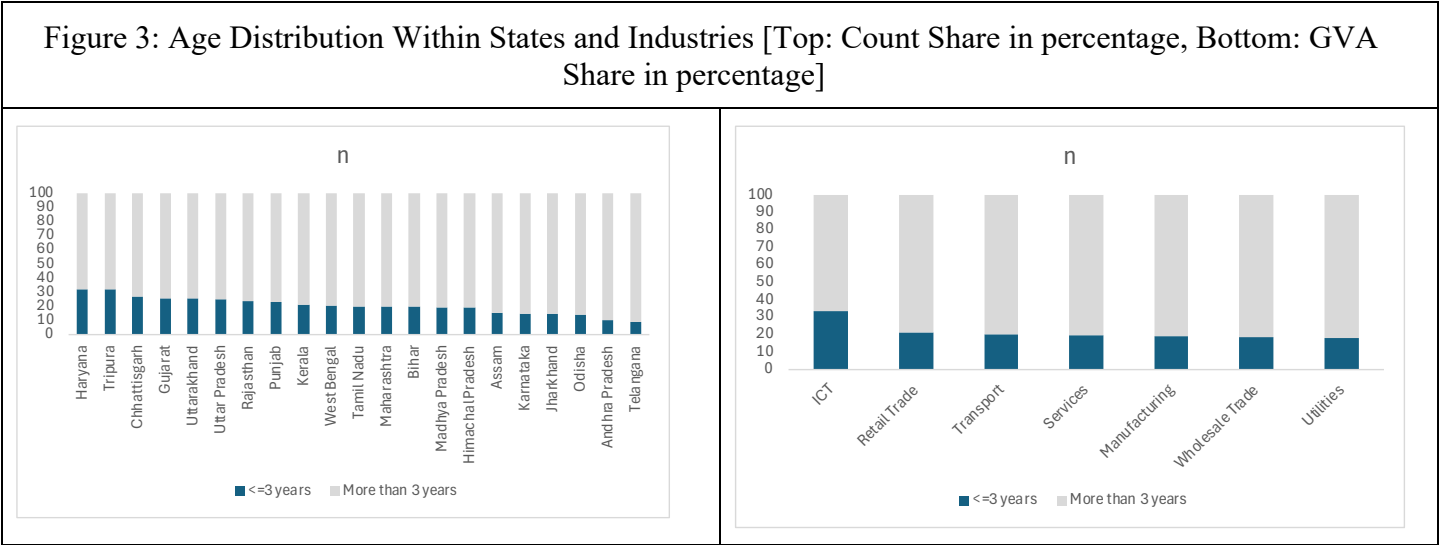
concentration of activity within a small number of industries also suggests that aggregate outcomes in the rural unincorporated sector are heavily influenced by the performance of a relatively narrow set of economic activities and policies targeting a small number of industries could have broad effects.

4. Skills x Industries



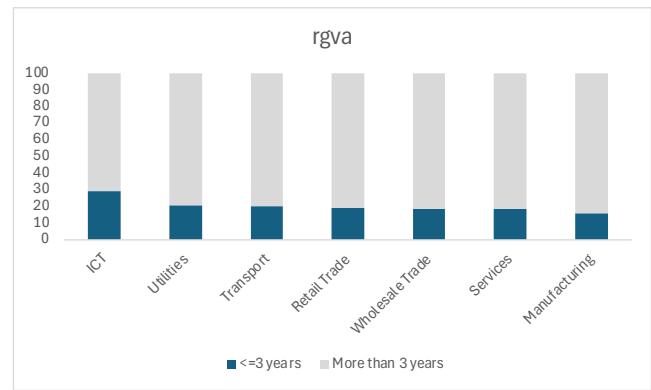
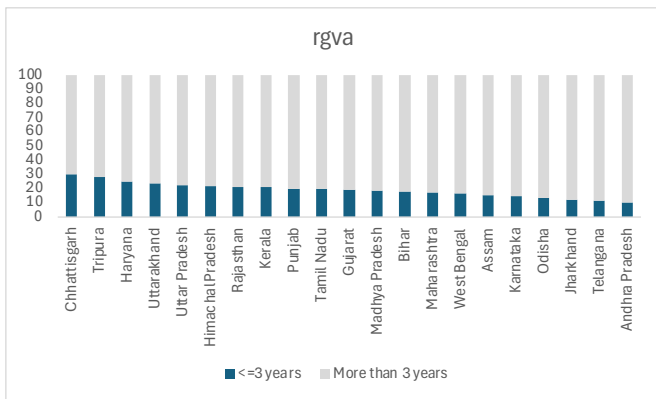
Skill composition differs considerably across industries. Most sectors are dominated by low-skill owners, but ICT stands out with a much larger presence of mid-skill¹⁷ owners in both establishment count and RGVA. This pattern may reflect the greater importance of educational attainment and technical skills in ICT-related activities compared to more traditional sectors.

5. Firm Age



¹⁷ For regression analysis (Section VI), mid-skill and high-skill category have been merged and is referred to as mid skill.

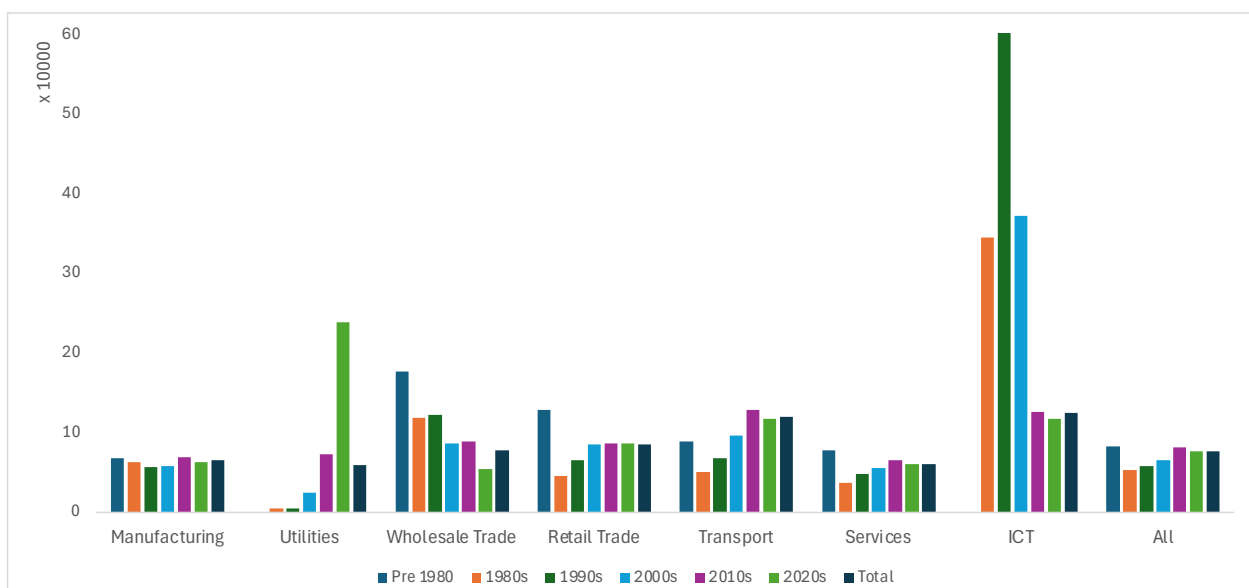
Figure 3: Age Distribution Within States and Industries [Top: Count Share in percentage, Bottom: GVA Share in percentage]



The relatively larger presence of newer firms in sectors such as ICT indicates greater recent entry into these activities, while in states such as Chhattisgarh and Tripura, younger firms account for a larger share of output relative to their share in establishments, suggesting that the contribution of new firms to output varies considerably across sectors and regions.

6. Year Cohorts

Figure 4: Median Investment Value across Industries by Cohorts (in ₹)



Note: Cohorts are based on their initial operation year. However, the median investments values are for the year 2023-24.

Median real fixed capital differs substantially across industries, pointing to important differences in enterprise scale and capital requirements. Sectors such as ICT, wholesale trade, and transport exhibit relatively higher investment levels compared to manufacturing and other services, though cohort patterns vary considerably across activities. For instance, ICT records markedly higher fixed capital among older enterprises, while more recent cohorts operate at substantially lower investment levels, suggesting that newer entrants may be able to operate with lower initial capital requirements than earlier cohorts.

V. Methodology

The empirical analysis uses cross-sectional data to analyse determinants of firm performance in terms of the real gross value added. The benchmark model¹⁸ is as follows:

$$\log RGVA_i = \beta_0 + \beta_1 \log Worker_i + \beta_2 \log FixedCapital_i + \beta_3 FirmCharacteristics_i + \beta_4 IndustryControls + \delta_j + \varepsilon_i$$

The dependent variable is the natural logarithm of real gross value added. The variable $\log(Worker)$ ¹⁹ gives the employment size and the $\log(fixed\ capital)$ gives the natural logarithmic value of (real) fixed capital²⁰. The firm characteristics consist of three variables: age of firm, if the firm uses technology²¹ or not, and if the firm is situated within household premises or not.

Industry fixed effects are included to account for differences in productivity, technology and market conditions across industries.

The model is further extended by including owner characteristics and policy variables that can influence the performance of the enterprises. Following is the extended model:

$$\log RGVA_i = \beta_0 + \beta_1 \log Worker_i + \beta_2 \log FixedCapital_i + \beta_3 FirmCharacteristics_i + \beta_4 OwnerCharacteristics_i + \beta_5 PolicyVariables_i + \beta_6 IndustryControls + \delta_j + \varepsilon_i$$

The owner characteristics include details of the owners such as, education level²² and gender of the owner. The policy variables include government expenditure, infrastructure index, and the governance index.

The *GovExpCat* variable classifies districts according to whether the change in government expenditure between 2022–23 and 2023–24 was negative, zero, or positive.

The infrastructure index is constructed using principal component analysis (PCA)²³ on Mission Antyodaya indicators covering road connectivity, bank access, market access, electricity availability, and piped water connections. The governance index is also constructed using PCA on Panchayat Advancement Index indicators related to local governance quality and women-friendly institutions.

In addition to the baseline specification, the model is estimated under alternative sample restrictions to examine heterogeneity in the estimated associations. Separate regressions are run for different skill groups and broad industry sectors. This allows the analysis to assess whether the relationship between enterprise-level RGVA and the explanatory variables varies across different types of enterprises and sectoral contexts.

Since the dependent variable is in logarithmic form, coefficients on logarithmic explanatory variables can be interpreted as elasticities. Coefficients on dummy variables represent approximate percentage differences relative to the omitted category. The estimates should be interpreted as conditional associations rather than causal effects, as unobserved firm- or district-level factors may still influence enterprise performance.

VI. Regression Results

¹⁸ For all the estimations, survey weights have been used.

¹⁹ $\log(Worker)$ is defined as $\log(\text{number of workers employed} + 1)$. One is added to the worker count prior to logarithmic transformation because a large proportion of enterprises employ only one worker, thereby avoiding a value of zero for these observations.

²⁰ Includes value of land, building, plant and machinery, transport equipment, furniture & fixtures, software and database, ICT equipment, tools and other fixed assets, and capital work in progress.

²¹ Technology variable indicates if the firm uses computer/internet.

²² Education Level variable has 5 categories: Not literate, Primary to Below Secondary, Secondary to below High Secondary, Higher secondary to Below Graduate, Graduate and above.

²³ PCA transforms potentially correlated variables into a smaller set of variables called Principal Components. This reduces the number of dimensions in large datasets while retaining most of the original information.

This section presents regression results (Table 3 & 4) from the models discussed in the previous section. Following table shows results from the benchmark model.

Table 3: Regression Results for Model 1	
<i>Dependent variable: Log of (Real) Gross Value Added</i>	
Independent variables	(1) All
Employment	1.0181*** (.137)
Log fixed capital	0.159*** (.019)
Firm age	0.025*** (0.004)
Firm age squared	-0.001*** (0.000)
Technology Use	0.299*** (0.028)
Within-household enterprise	-0.595*** (0.061)
Constant	8.203*** (0.215)
Observations	35,252,323
R-squared	0.4913
Within R-squared	0.2919
Notes: Standard errors clustered at the two-digit NIC industry level are reported in parentheses. Survey weights are used in all specifications. The reference categories are no computer/internet use; enterprise located outside the household. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.	

The results indicate that labour and fixed capital are positively associated with enterprise-level RGVA. Firm age displays an inverted-U relationship with RGVA, implying that enterprise performance improves with experience but at a diminishing rate. Technology-using enterprises (i.e. using computer or internet) report significantly higher RGVA, which might reflect improved access to information, customers, and business processes among digitally connected enterprises. However, enterprises operating from within the household report substantially lower RGVA, consistent with the idea that home-based enterprises operate at a smaller scale and face greater constraints in expansion and market access.

Table 4: Regression Results for Model 2*Dependent variable: Log of (Real) Gross Value Added*

Independent variables	(1) All	(2) Low skill	(3) Mid skill	(4) Manufacturing	(5) Trade & Other Services
Employment	0.580*** (0.092)	0.450*** (0.130)	0.678*** (0.069)	0.648*** (0.055)	0.525*** (0.124)
Log fixed capital	0.159*** (0.013)	0.162*** (0.017)	0.146*** (0.014)	0.235*** (0.015)	0.130*** (0.013)
Firm age	0.020*** (0.003)	0.015*** (0.003)	0.024*** (0.004)	0.022*** (0.007)	0.019*** (0.003)
Firm age squared	-0.001*** (0.000)	-0.000*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Technology Use	0.256*** (0.016)	0.311*** (0.023)	0.215*** (0.015)	0.298*** (0.069)	0.256*** (0.016)
Within-household enterprise	-0.406*** (0.030)	-0.399*** (0.032)	-0.386*** (0.035)	-0.568*** (0.061)	-0.354*** (0.033)
Education of the Owner					
Primary to below secondary	0.123*** (0.030)	0.114*** (0.029)	-0.022 (0.068)	0.107** (0.047)	0.135*** (0.037)
Secondary to below higher secondary	0.200*** (0.041)	— —	-0.048 (0.067)	0.165*** (0.047)	0.225*** (0.053)
Higher secondary to below graduate	0.244*** (0.065)	— —	-0.004 (0.075)	0.137** (0.053)	0.304*** (0.067)
Graduate and above	0.375*** (0.089)	— —	0.119 (0.089)	0.199*** (0.057)	0.457*** (0.085)
Female owner	-0.352*** (0.059)	-0.359*** (0.072)	-0.298*** (0.038)	-0.638*** (0.118)	-0.278*** (0.028)
Female × Within-household	-0.351*** (0.061)	-0.325*** (0.078)	-0.406*** (0.056)	-0.105 (0.092)	-0.294** (0.138)
Policy Variables					

Table 4: Regression Results for Model 2*Dependent variable: Log of (Real) Gross Value Added*

Independent variables	(1) All	(2) Low skill	(3) Mid skill	(4) Manufacturing	(5) Trade & Other Services
Infrastructure Index	0.768*** (0.118)	0.845*** (0.183)	0.622*** (0.072)	0.719*** (0.259)	0.802*** (0.124)
GovExpCat: zero change	0.420*** (0.154)	0.594** (0.234)	0.128*** (0.045)	0.722*** (0.208)	0.338 (0.203)
GovExpCat: positive change	0.292* (0.166)	0.304 (0.224)	0.286** (0.114)	0.098 (0.350)	0.416*** (0.126)
Infrastructure × zero change	-0.334 (0.205)	-0.532* (0.316)	-0.004 (0.076)	-0.676** (0.327)	-0.276 (0.267)
Infrastructure × positive change	-0.538* (0.270)	-0.562 (0.350)	-0.526** (0.204)	-0.217 (0.576)	-0.738*** (0.199)
Governance index	-0.081 (0.051)	-0.165*** (0.059)	0.031 (0.040)	0.001 (0.096)	-0.117** (0.044)
Constant	8.504*** (0.141)	8.505*** (0.213)	8.887*** (0.151)	7.733*** (0.232)	8.835*** (0.183)
Observations	35,093,327	18,079,393	17,013,934	11,750,215	23,343,112
R-squared	0.559	0.509	0.587	0.580	0.414
Within R-squared	0.376	0.302	0.389	0.452	0.338
Clustered SE	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors clustered at the two-digit NIC industry level are reported in parentheses. Survey weights are used in all specifications. The reference categories are no ICT use, male owner, enterprise located outside the household, no education, and <0 GovExpCat²⁴ category. *, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.

Column (1) reports the baseline specification that includes all observations. Columns (2) and (3) present estimates for low- and mid-skill enterprises, while Columns (4) and (5) present estimates for manufacturing and trade/other services enterprises, respectively. The results suggest that enterprise scale

²⁴ GovExpCat has 3 categories: <0, =0 and >0 showing the direction of change in district level development expenditure from 2022-23 to 2023-24.

(in terms of investment and employment size), capital intensity and technology adoption are strongly associated with higher enterprise performance.

- a. **Firm Characteristics:** The coefficient on log workers is positive and statistically significant across all specifications, indicating that larger enterprises report higher RGVA. The association is stronger for mid-skill enterprises than for low-skill enterprises. Similarly, the coefficient on log fixed capital is positive and significant throughout, with the largest effect observed in manufacturing enterprises. Results for other variables; firm age, technology use, location of the firm remain similar to the benchmark model, across all specifications.
- b. **Owner Characteristics:** In the baseline specification, higher levels of owner education are associated with higher RGVA relative to the uneducated category. The coefficients generally increase with education level, suggesting a clear education gradient in enterprise performance. This pattern is particularly visible among trade and service enterprises. The results are broadly consistent with the view that education improves managerial ability, access to information, and the adoption of productive technologies.

Female-owned enterprises report significantly lower RGVA than male-owned enterprises across all specifications, conditional on other covariates. The magnitude of this difference is especially large in manufacturing, suggesting that gender-related constraints may be more pronounced in this sector. The interaction between female ownership and within-household operation is negative in all specifications and statistically significant in most cases, indicating an additional disadvantage for enterprises that are both female-owned and home-based, beyond the separate effects of gender and enterprise location.

- c. **District-Level Characteristics:** The infrastructure index is positive and statistically significant across all specifications, indicating that enterprises located in districts with better infrastructure conditions tend to report higher RGVA. The interaction terms between infrastructure and government expenditure categories suggest that this relationship varies with changes in district-level developmental expenditure between 2022–23 and 2023–24.

The negative interaction coefficients imply that the positive association between infrastructure and RGVA is weaker in districts experiencing zero or positive expenditure growth relative to districts with declining expenditure, although the statistical significance of these effects varies across specifications. In contrast, the governance index does not exhibit a consistent association with RGVA. This may suggest that infrastructure-related factors have a more direct relationship with enterprise performance, while the effects of governance are either weaker or less effectively captured by the available district-level indicators.

The regression results suggest that enterprise performance in the rural unincorporated sector is closely linked to enterprise scale, capital availability, human capital, digital adoption and local infrastructure conditions. Enterprises with more workers, higher fixed capital, better educated owners and ICT usage report higher RGVA, while female-owned and home-based enterprises report lower RGVA. The results also indicate heterogeneity across skill groups and sectors, with capital playing a stronger role in manufacturing and the education association being more visible in trade and other services.

VII. Conclusion

This study examines the factors determining the performance of enterprises in the Indian rural unincorporated sector. It highlights the role of enterprise and ownership characteristics such as firm age, education of the owner, technology adoption and the external environment, particularly fiscal and infrastructure support, in shaping outcomes.

The rural unincorporated sector is marked by substantial heterogeneity across sectors, industries, states, entrepreneurial skills, and enterprise age cohorts. Descriptive patterns indicate that enterprise activity is concentrated in a small number of industries, while sectoral differences in GVA, fixed capital, and skill composition point to important variation in the scale and nature of rural enterprise activity.

The regression results further show that enterprise performance is positively associated with employment size, fixed capital, education qualifications of the owner, ICT use, and district-level infrastructure. At the same time, female-owned enterprises and enterprises operating from within the household report lower RGVA, suggesting that ownership and location-based constraints remain important within this sector. The findings also reveal that enterprise maturity matters, but only up to a point, as RGVA appears to rise with firm age in the initial years before declining among older establishments. These associations vary across skill groups and sectors, with capital playing a stronger role in manufacturing and the gains from education being more visible in trade and other services.

Strengthening the rural unincorporated sector requires policies that recognise that different enterprises face different constraints. Many rural unincorporated enterprises operate at a scale far below the conventional definition of MSMEs and are better characterised as nano-enterprises, suggesting that policies designed for the broader MSME sector may not adequately address their constraints. For female-owned and home-based establishments, this may involve improving access to finance, markets, and opportunities for expansion, while greater ICT adoption and improvements in local infrastructure can support productivity growth across the sector.

While the present study focuses on the rural sector, further research could extend the analysis to urban areas to examine whether the observed relationships persist across different economic and institutional contexts. Such an extension would also enable a comparative assessment of rural-urban development dynamics and their policy implications.

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